

| Question | Answer                                                                                                                                                                            | Marks       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 4(a)     | random/haphazard                                                                                                                                                                  | <b>B1</b>   |
|          | constant velocity <b>or</b> speed in a straight line between collisions<br><b>or</b><br>distribution of speeds/different directions                                               | <b>B1</b>   |
| 4(b)     | (small) specks of light/bright specks/pollen grains/dust particles/smoke particles                                                                                                | <b>M1</b>   |
|          | moving haphazardly/randomly/jerky/in a zigzag fashion                                                                                                                             | <b>A1</b>   |
| 4(c)(i)  | $pV = \frac{1}{3} Nm\langle c^2 \rangle$<br>$1.05 \times 10^5 \times 0.0240 = \frac{1}{3} \times 4.00 \times 10^{-3} \times \langle c^2 \rangle$                                  | <b>C1</b>   |
|          | $\langle c^2 \rangle = 1.89 \times 10^6$                                                                                                                                          | <b>C1</b>   |
|          | <b>or</b>                                                                                                                                                                         |             |
|          | $\frac{1}{2} m\langle c^2 \rangle = (3/2) kT$<br>$0.5 \times (4.00 \times 10^{-3} / 6.02 \times 10^{23}) \times \langle c^2 \rangle = 1.5 \times 1.38 \times 10^{-23} \times 300$ | <b>(C1)</b> |
|          | $\langle c^2 \rangle = 1.87 \times 10^6$                                                                                                                                          | <b>(C1)</b> |
|          | <b>or</b>                                                                                                                                                                         |             |
|          | $nRT = \frac{1}{3} Nm\langle c^2 \rangle$<br>$1.00 \times 8.31 \times 300 = \frac{1}{3} \times 4.00 \times 10^{-3} \times \langle c^2 \rangle$                                    | <b>(C1)</b> |
|          | $\langle c^2 \rangle = 1.87 \times 10^6$                                                                                                                                          | <b>(C1)</b> |
|          | $c_{r.m.s.} = 1.37 \times 10^3 \text{ m s}^{-1}$                                                                                                                                  | <b>A1</b>   |

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|----------|-------------------------------------------------------------------------|-----------|
| 4(c)(ii) | $\langle c^2 \rangle \propto T$                                         | <b>C1</b> |
|          | $\langle c^2 \rangle$ at 177 °C = $1.89 \times 10^6 \times (450 / 300)$ | <b>C1</b> |
|          | $c_{\text{r.m.s.}}$ at 177 °C = $1.68 \times 10^3 \text{ m s}^{-1}$     | <b>A1</b> |

| Question | Answer                                                                                                                                                   | Marks     |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)     | (loss in) kinetic energy of $\alpha$ -particle = $Qq / 4\pi\epsilon_0 r$<br><b>or</b><br>$7.7 \times 10^{-13} = Qq / 4\pi\epsilon_0 r$                   | <b>C1</b> |
|          | $7.7 \times 10^{-13} = 8.99 \times 10^9 \times 79 \times 2 \times (1.60 \times 10^{-19})^2 / r$                                                          | <b>M1</b> |
|          | $r = 4.7 \times 10^{-14} \text{ m}$<br>$r$ is closest distance of approach so radius less than this                                                      | <b>A1</b> |
| 5(b)     | force = $Qq / 4\pi\epsilon_0 r^2 = 4u \times a$                                                                                                          | <b>C1</b> |
|          | $8.99 \times 10^9 \times 79 \times 2 \times (1.60 \times 10^{-19})^2 / (4.7 \times 10^{-14})^2 = 4 \times 1.66 \times 10^{-27} \times a$                 | <b>C1</b> |
|          | $a = 2.5 \times 10^{27} \text{ m s}^{-2}$                                                                                                                | <b>A1</b> |
| 5(c)     | so that single interactions between nucleus and $\alpha$ -particle can be studied<br><b>or</b><br>so that multiple deflections with nucleus do not occur | <b>B1</b> |